

Modern Physics

Lecture 3: X rays and Diffraction

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Books: *Ch 2 of* Arthur Beiser,
Concepts of Modern Physics
Ch 38.2 of Hugh D. Young and Roger A.
Freedman, University Physics with Modern Physics

Recap: what have we learned about matter and radiation?

Photoelectric Effect

Electromagnetic radiation falls on a metal surface and electrons are emitted.

Wave picture: Electrons should be emitted when an electromagnetic wave of sufficiently large intensity falls on the metal, because a more intense wave carries more energy and should eventually provide enough energy for electrons to overcome the work function.

Experimentally, we found that electrons are emitted only when the light frequency exceeds a *threshold frequency*, irrespective of how intense the light is.

- Above this threshold, increasing the frequency increases the maximum kinetic energy of the emitted electrons.
- Increasing the intensity mainly increases the number of emitted electrons.

Recap: what have we learned about matter and radiation?

Based on Einstein's explanation, electromagnetic radiation exchanges energy with matter in discrete packets called photons, each carrying energy $h\nu$.

$$\begin{aligned} \text{Photon} + \text{electron in metal} &\implies \text{free electron} \\ h\nu &= \phi + \text{Kinetic Energy of electron} \end{aligned}$$

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$$h\nu = \phi + \text{Kinetic Energy of electron}$$

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Recap: what have we learned about matter and radiation?

The energy in EM wave or beam is determined by total no. of photons there are. So at a fixed frequency, increasing the intensity means increasing the number of photons arriving per second per unit area.

Both beams have the same frequency ν (same wavelength λ)
and hence the same photon energy $E = h\nu$.

Low intensity



Each photon: $E = h\nu$

Fewer photons per second
(in this example: 4 photons)

Energy of the electromagnetic wave (during the same time interval):

$$E_{\text{beam}} = 4h\nu$$

High intensity



Each photon: $E = h\nu$

More photons per second
(in this example: 10 photons)

Energy of the electromagnetic wave (during the same time interval):

$$E_{\text{beam}} = 10h\nu$$

Same frequency $\nu \rightarrow$ same photon energy $E = h\nu$ for each photon.
Higher intensity means a larger number of photons per second, and hence a larger total energy.

Assuming ideal monochromatic source of EM wave

Recap: what have we learned about matter and radiation?

Wave picture: Intensity $\propto E^2$ Photon picture: Intensity $\propto nh\nu$

$$u = \varepsilon_0 E^2 = \frac{B^2}{\mu_0} = n_1 h\nu_1 + n_2 h\nu_2 + \dots$$

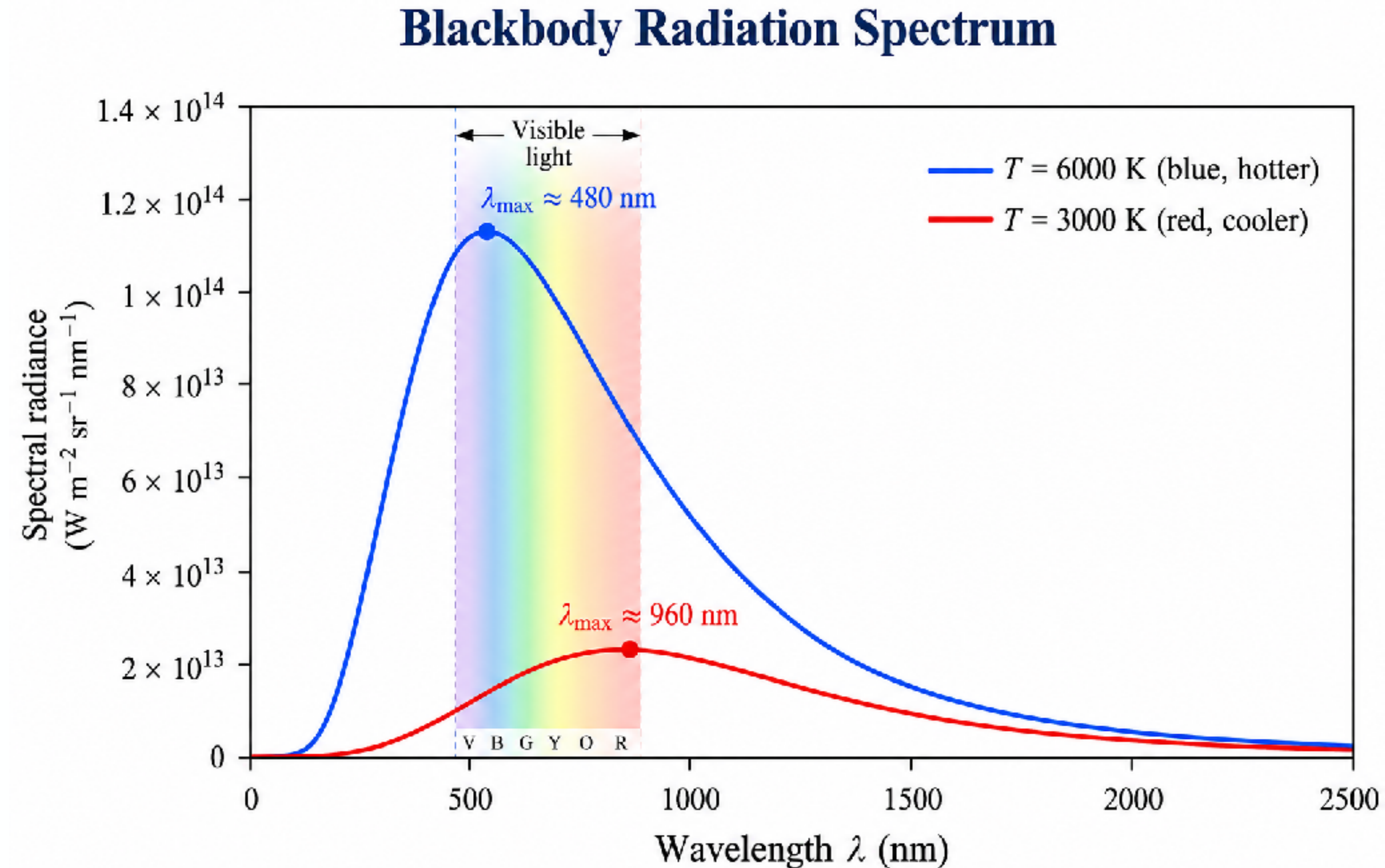
**More photons with
threshold frequency
there are, more
electrons absorb
photons and are
emitted from the metal.**

- ☑ Above this threshold, increasing the frequency increases the maximum kinetic energy of the emitted electrons.
- ☑ Increasing the intensity mainly increases the number of emitted electrons.

Recap: what have we learned about matter and radiation?

Black Body Radiation

We heated an object. Charges inside matter are in thermal motion and interact/accelerate, so the hot object emits electromagnetic radiation. Experimentally, when we separate this radiation according to wavelength, the intensity is not the same at every wavelength: we obtain the blackbody spectrum.



Recap: what have we learned about matter and radiation?

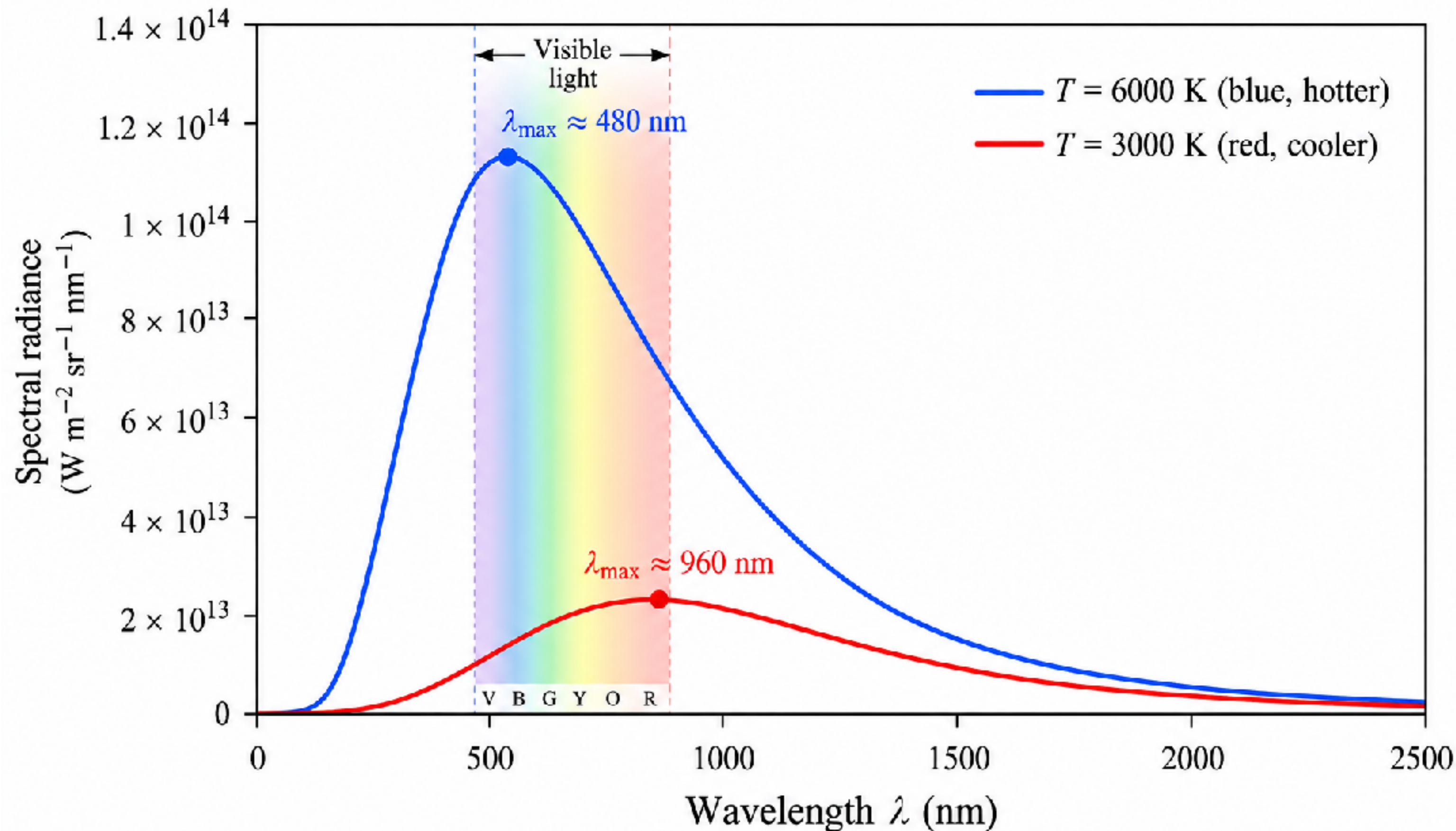
Temperature sets the thermal energy scale $\Rightarrow$ sets the characteristic photon frequency.

$$u = \varepsilon_0 E^2 = \frac{B^2}{\mu_0} = n_1 h\nu_1 + n_2 h\nu_2 + \dots$$

The number density of photons and frequency both decide the total energy of EM wave/beam.

Recap: what have we learned about matter and radiation?

Blackbody Radiation Spectrum



- Why does the characteristic frequency (colour) of emitted light increase with temperature?

Einstein's photon picture explains why the energy of light depends on frequency: higher-frequency photons carry more energy ($E = h\nu$).

- Shape

The distribution of photons will decide its shape.

Reverse of Photoelectric Effect

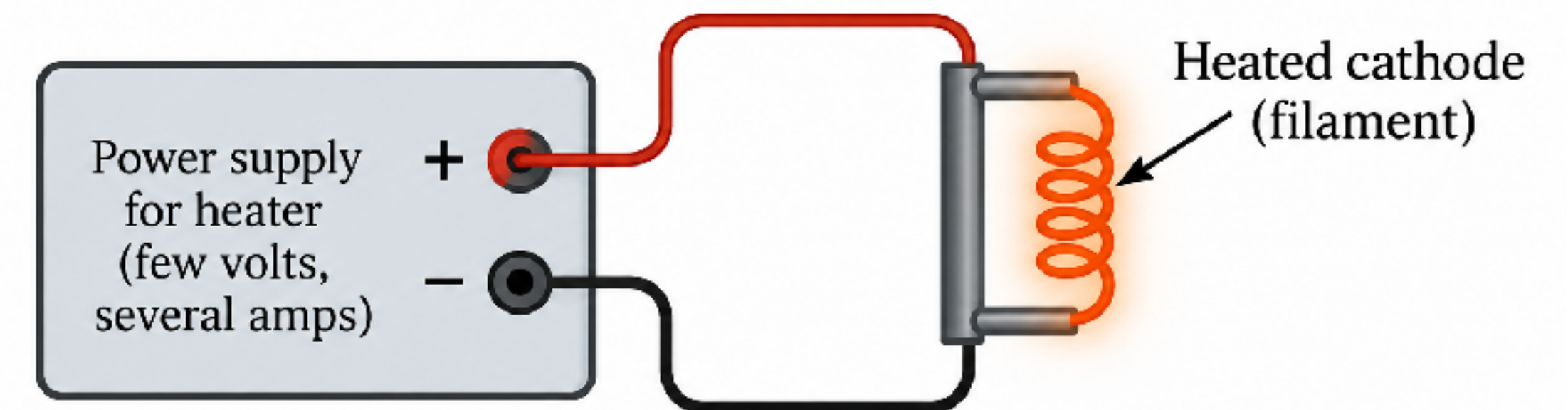
In blackbody radiation, matter was giving energy to electromagnetic radiation. So in a broad sense it already looks like the opposite direction from the photoelectric effect. However, if we want to design a clean quantitative experiment, we need to know the exact energy of the electrons and be able to control it.

Reverse of Photoelectric Effect

**Goal: I want to design a experiment where I can control energy of electrons. Then I want to make those electrons interact with matter and ask: what electromagnetic radiation do they produce?
How do I understand this in photon picture?**

Producing Electrons - Thermionic Emission

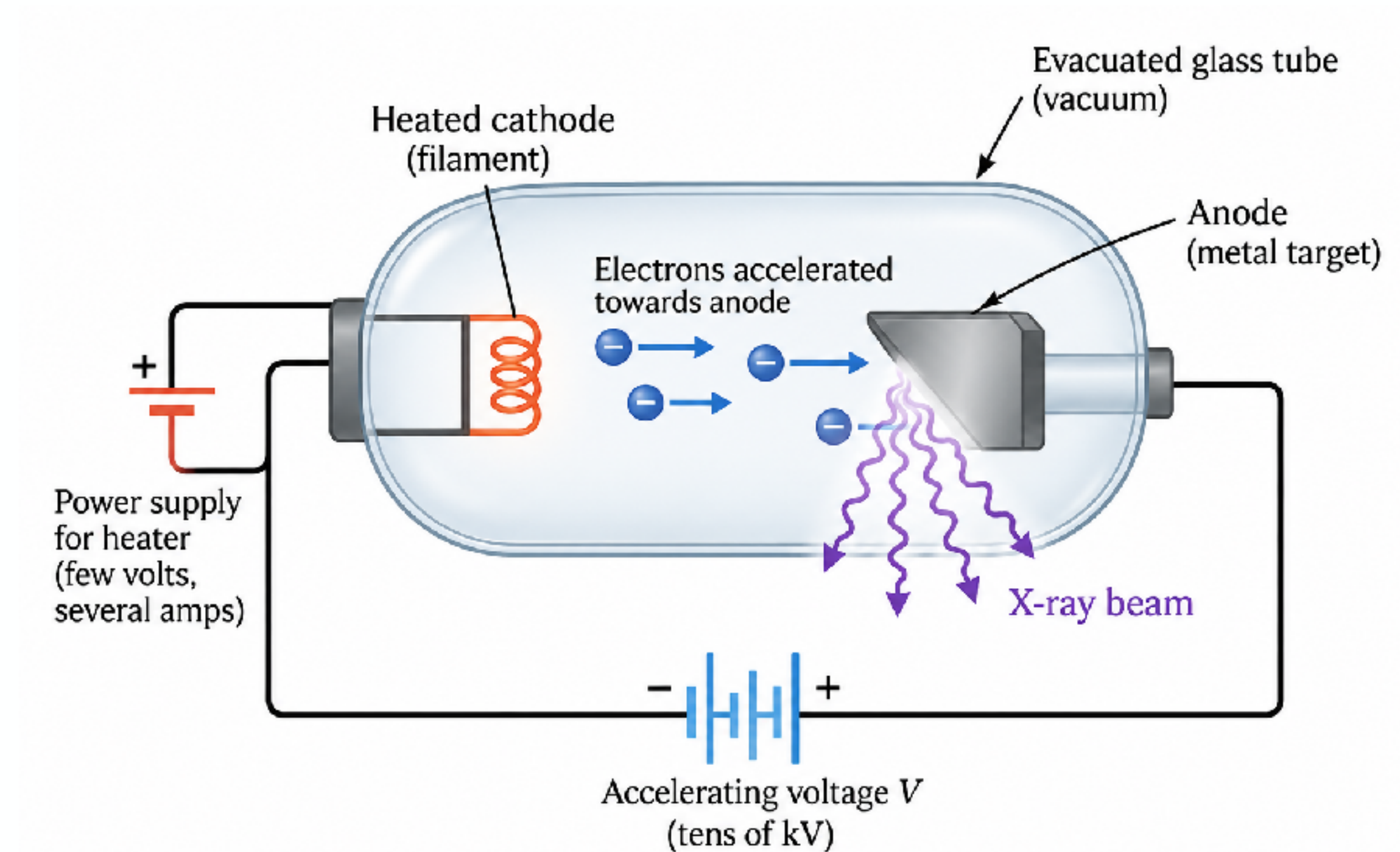
- First, we need **electrons outside the metal**, so we heat the cathode. Some electrons gain enough thermal energy to overcome the **work function** and escape from the metal. This process is called **thermionic emission**.
- Let me approximately treat these emitted electrons as starting from rest.



Circuit Diagram: Energy of electrons

Let's give each electron **known** energy.
I connect the cathode and anode to a high power supply. Suppose $V = 50 \text{ kV}$.

The electric field exerts a force on the electron:
 $-e\mathbf{E}$.



Circuit Diagram: Energy of electrons

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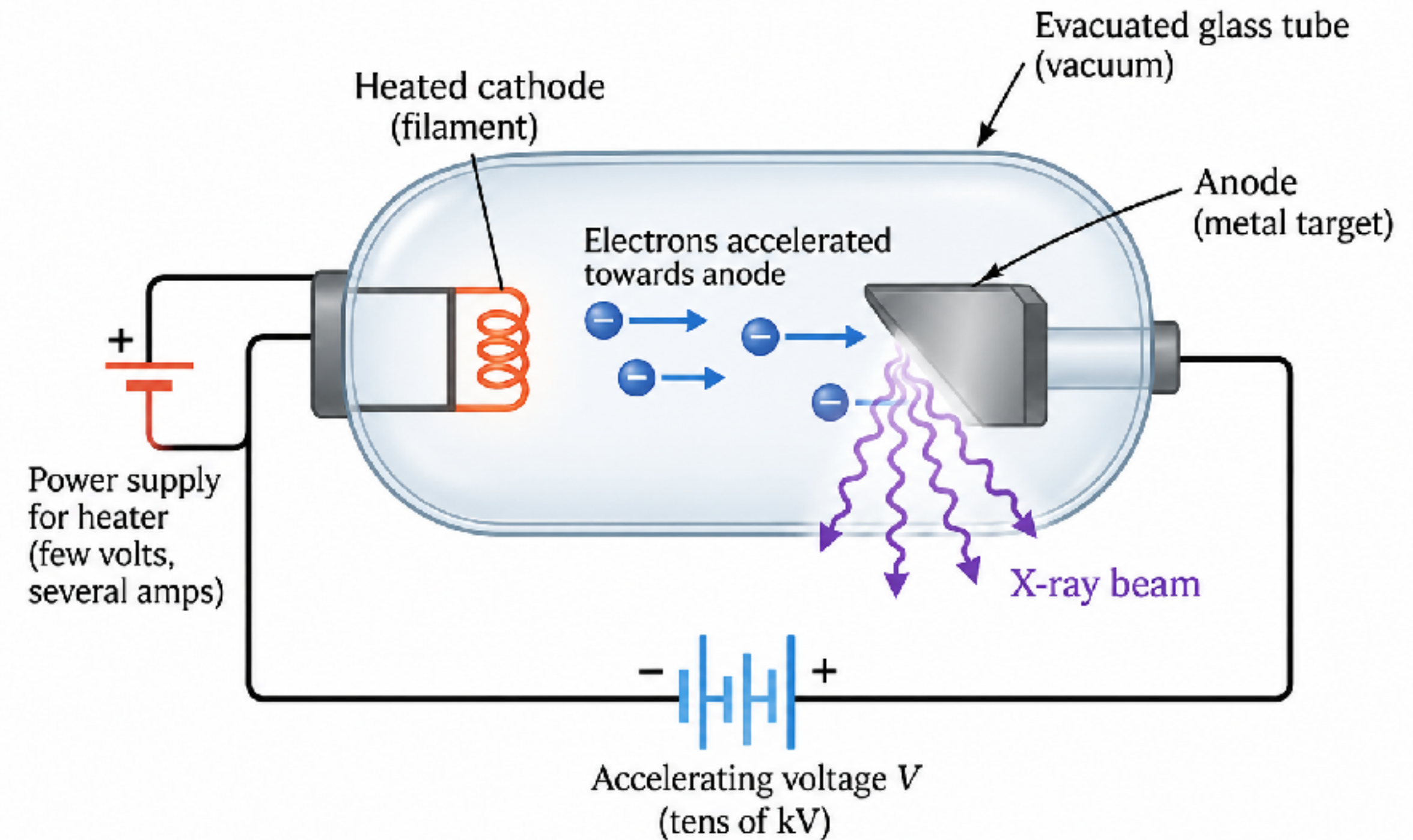
Therefore, it accelerates toward the positive anode.
Now use work-energy:

$$\Delta K = -\Delta U$$

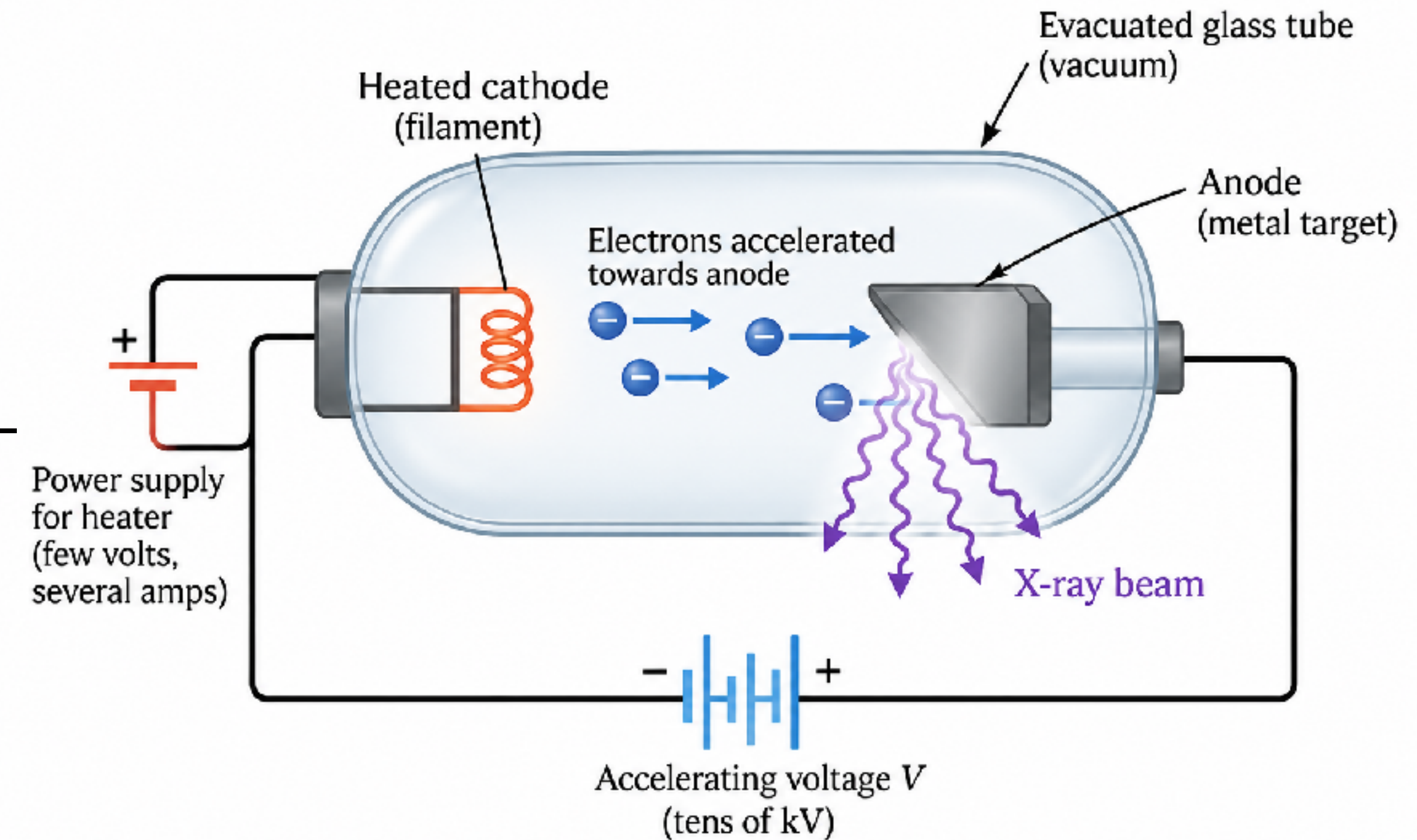
For an electron crossing a potential difference,

$$K = eV = 50 \text{ keV}$$

**This 50 keV did not come from heating the filament.
It came from the high-voltage power supply.**

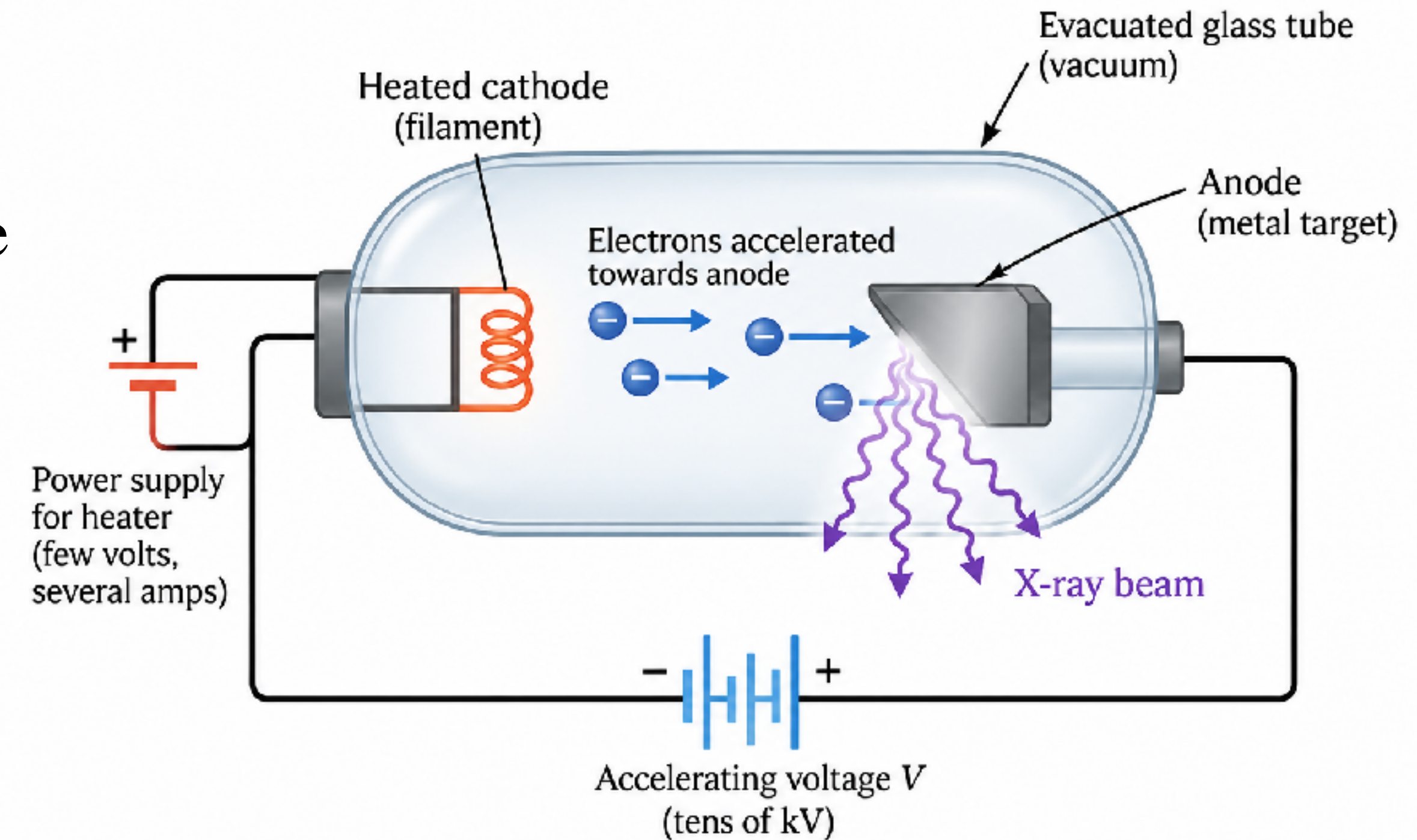


What happens when these (50 keV) electrons reach the anode? Just as in the circuits we have encountered earlier—for example, in the photoelectric-effect experiment—the electrons are collected by the anode and **constitute an electric current**.



However, something else can happen because these electrons are highly energetic. They can penetrate into the material of the anode and pass close to its atomic nuclei.

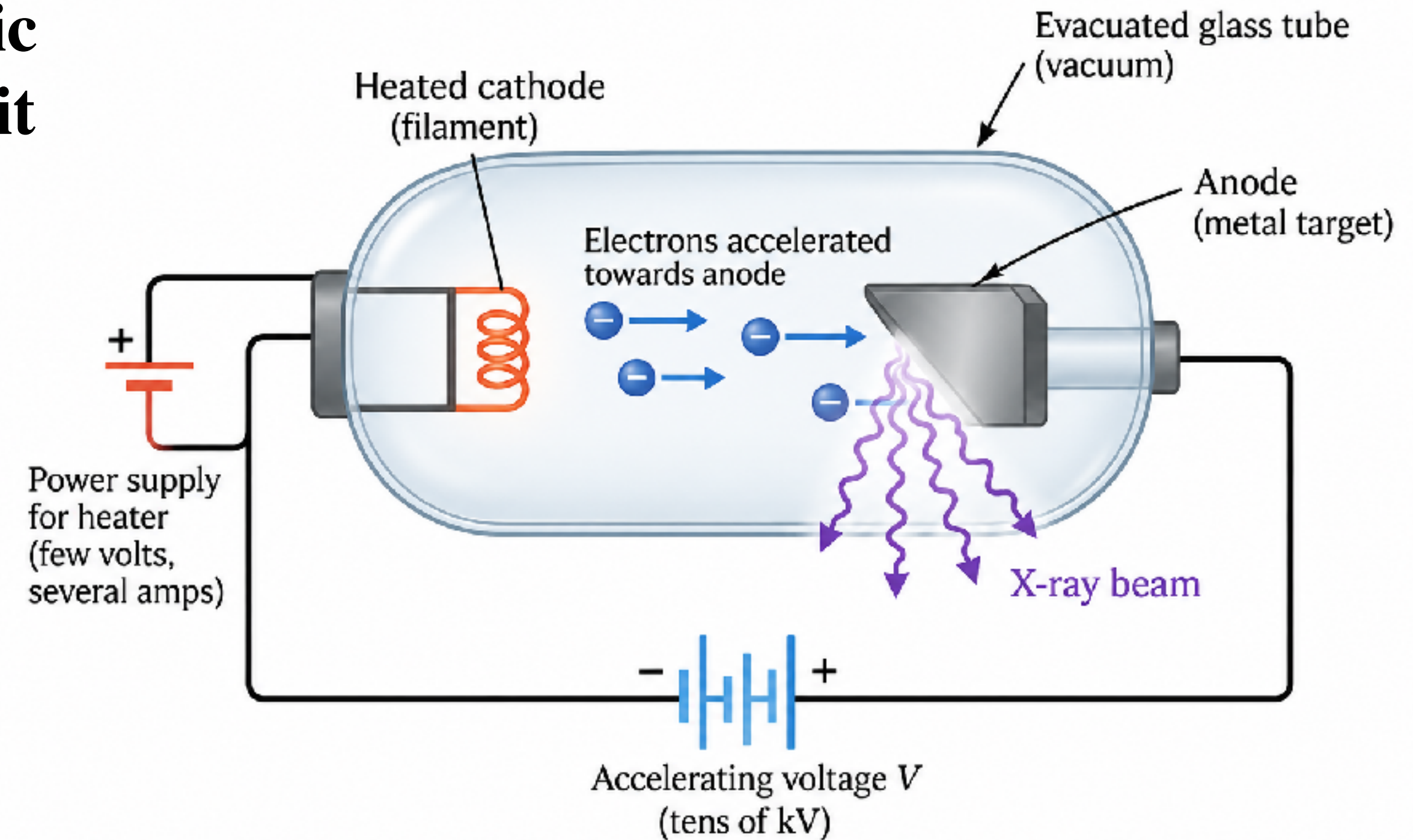
When an electron passes close to a positively charged nucleus, the electric field of the nucleus attracts it and **bends its trajectory**. Therefore, the velocity of the electron changes. As it moves away from the nucleus, the attractive force can also slow it down.



Bremstrahlung Radiation

An accelerating charged particle emits electromagnetic radiation. Therefore, these energetic electrons can emit electromagnetic radiation while interacting with the nuclei of the anode.

Based on Einstein's explanation of the photoelectric effect, we now interpret electromagnetic radiation as being made of photons.



X ray production

Based on Einstein's explanation of the photoelectric effect, we now interpret electromagnetic radiation as being made of photons.

What is the maximum possible energy of one photon emitted by a 50 keV electron?

$$\text{K.E of electron} = 50 \text{ keV} = h\nu_{\text{max}}$$

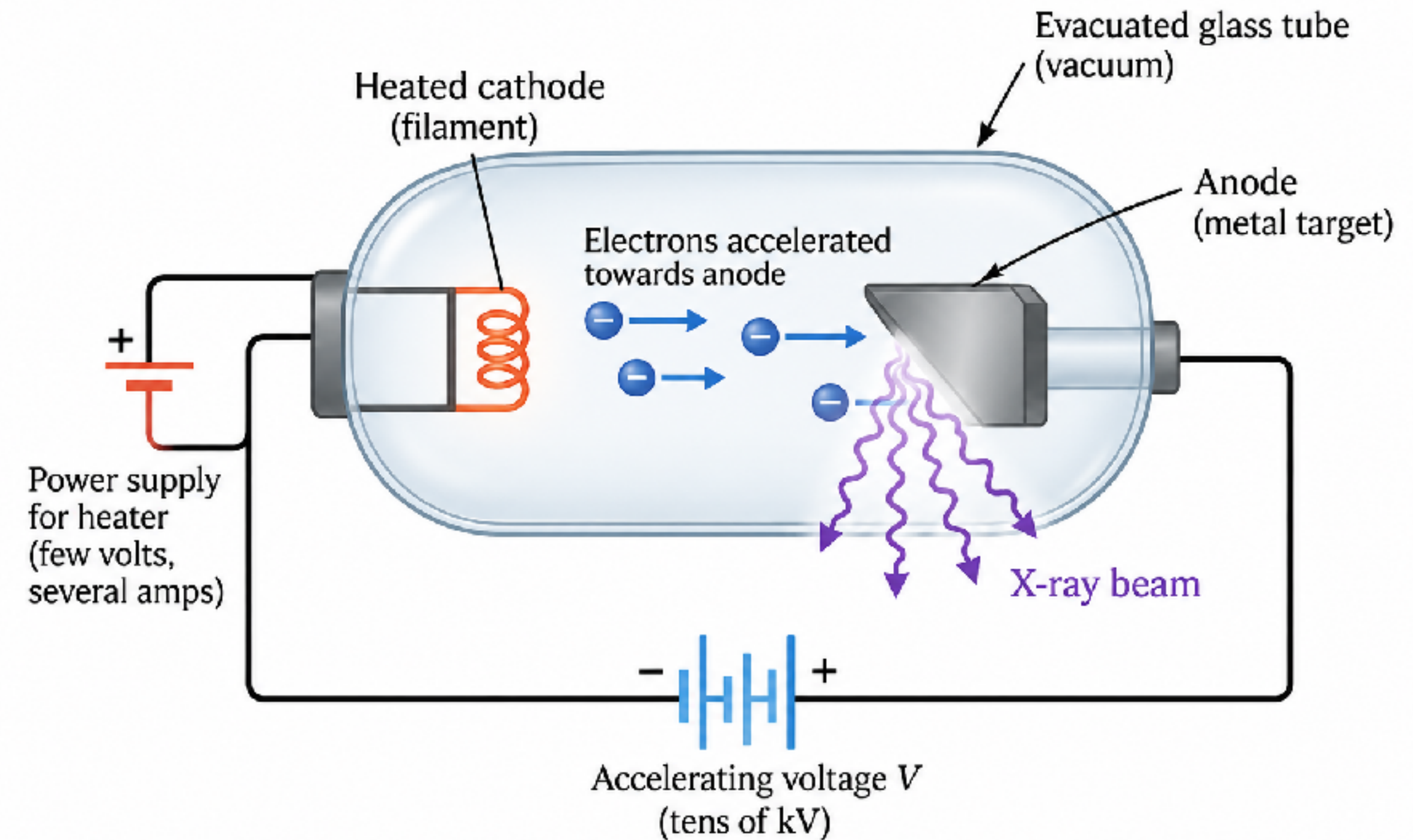
$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

$$h = 6.626 \times 10^{-34} \text{ Js}$$

$$\nu_{\text{max}} = \frac{\text{K.E}}{h} \approx 1.2 \times 10^{19} \text{ Hz}$$

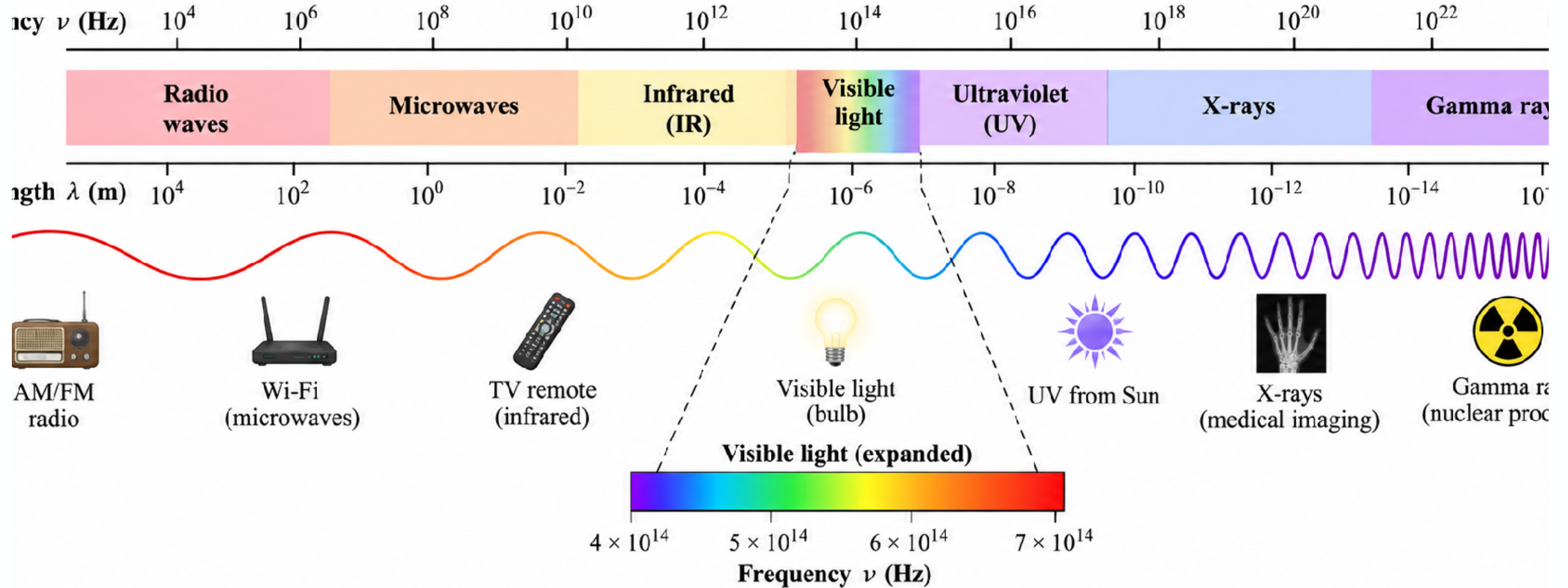
$$\lambda_{\text{min}} = 0.25 \text{ \AA}$$

X-Rays



X ray production

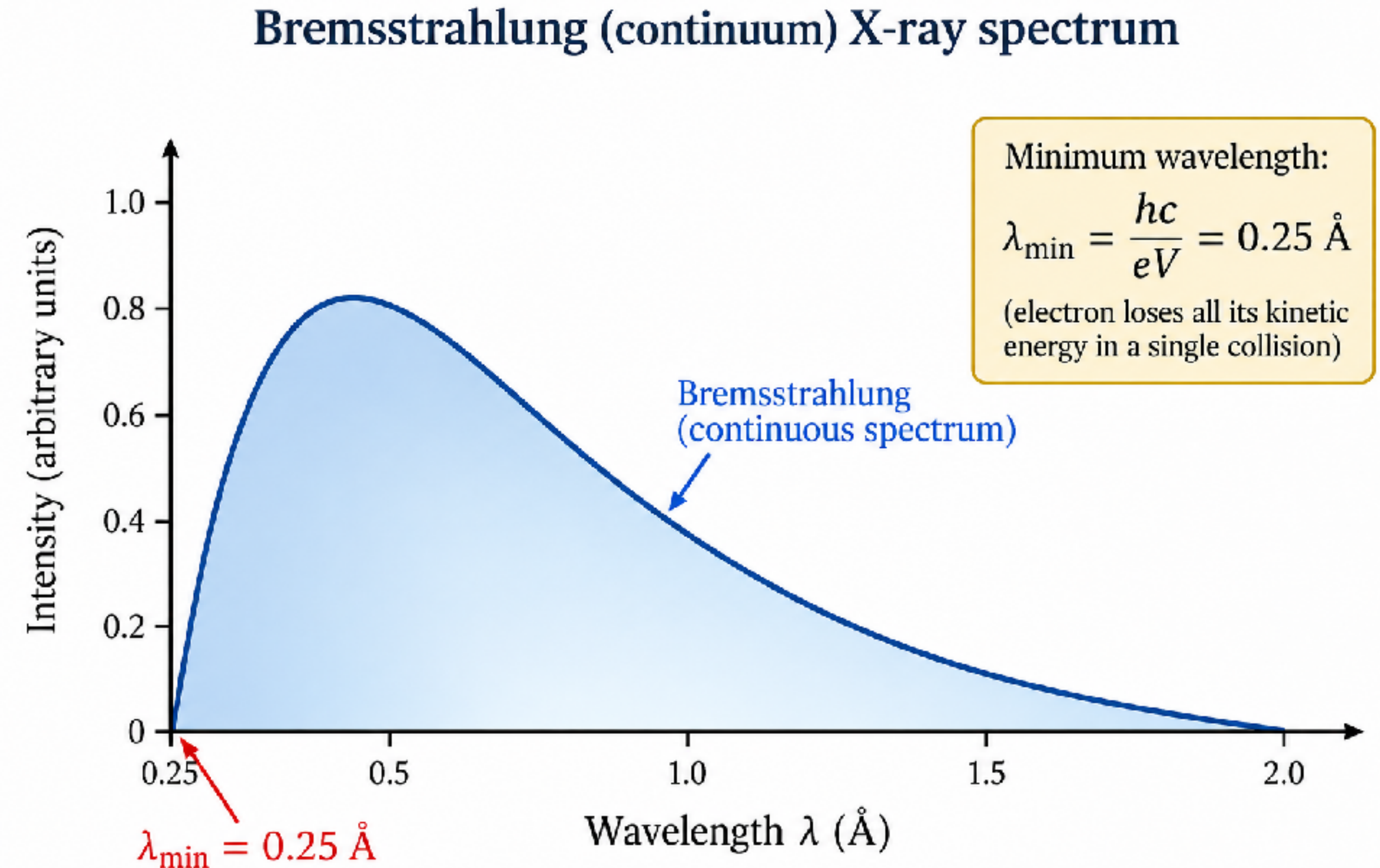
The Electromagnetic Spectrum



Observation

Now let us perform the experiment. We place a detector outside the tube and measure the electromagnetic radiation coming from the target. **We indeed observe X-rays.**

But we do **not** observe only photons of 50 keV. An electron does not usually transfer all of its kinetic energy to a single photon. For example, it might emit a 5 keV photon and retain approximately 45 keV, or it might emit a 20 keV photon and retain approximately 30 keV.



Additional EM radiation

* Does the heated cathode itself emit electromagnetic radiation?

Yes. When we heat the tungsten filament, the charged particles inside the material are in thermal motion and acceleration, so the filament also emits thermal electromagnetic radiation, just like the blackbody radiation discussed earlier - that is why the filament glows.

* What about the electrons while they are being accelerated from cathode to anode?

The electrons are accelerated by the electric field throughout the tube, so in principle an accelerating charge can radiate even before reaching the anode. However, this radiation is extremely weak compared with the radiation produced when the electrons encounter the very strong and rapidly varying electric fields near the nuclei in the anode.

Revisiting Diffraction

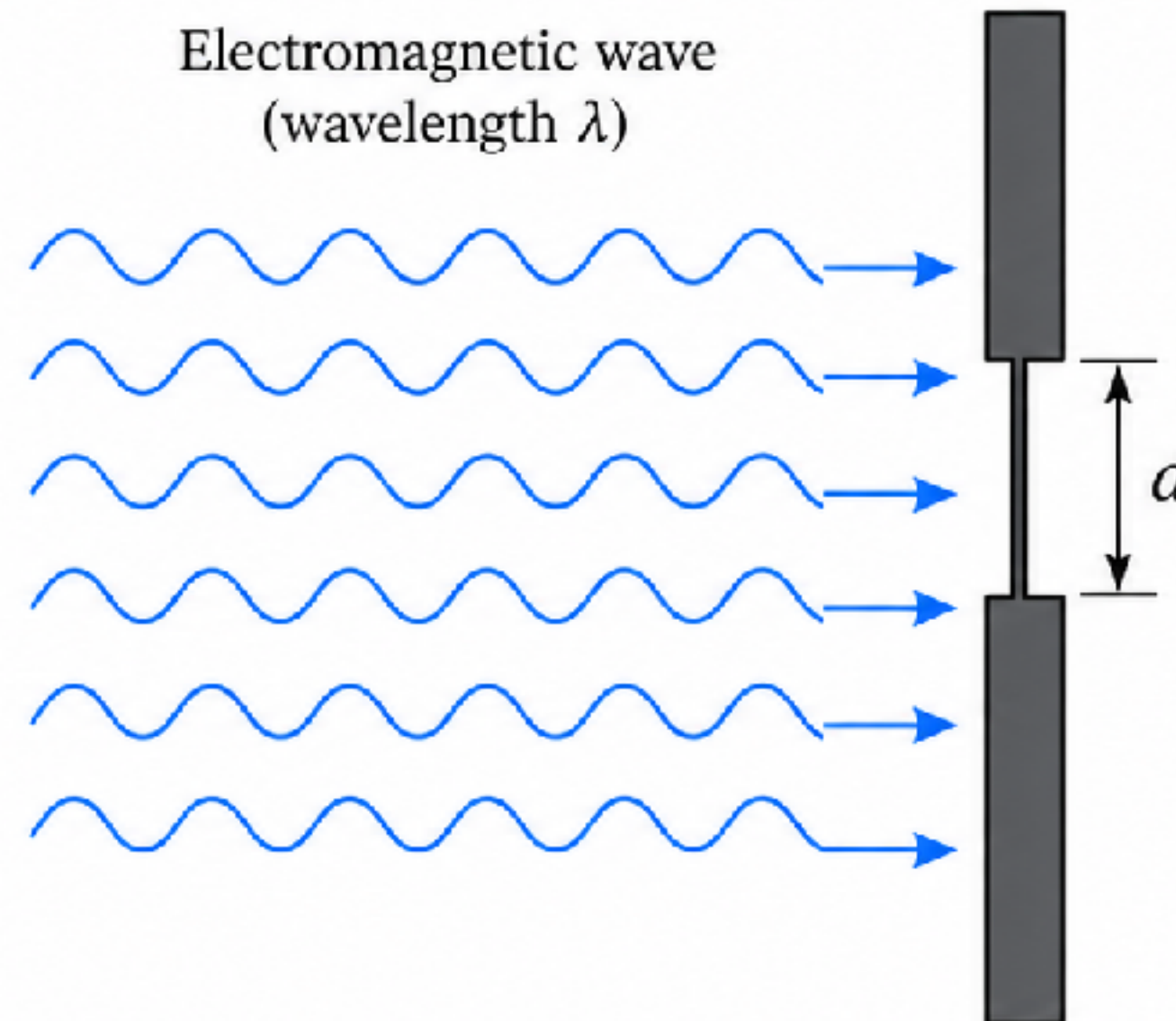
Question: If electromagnetic radiation consists of photons, what happens to the photons in a diffraction or interference experiment?

Revisiting Diffraction

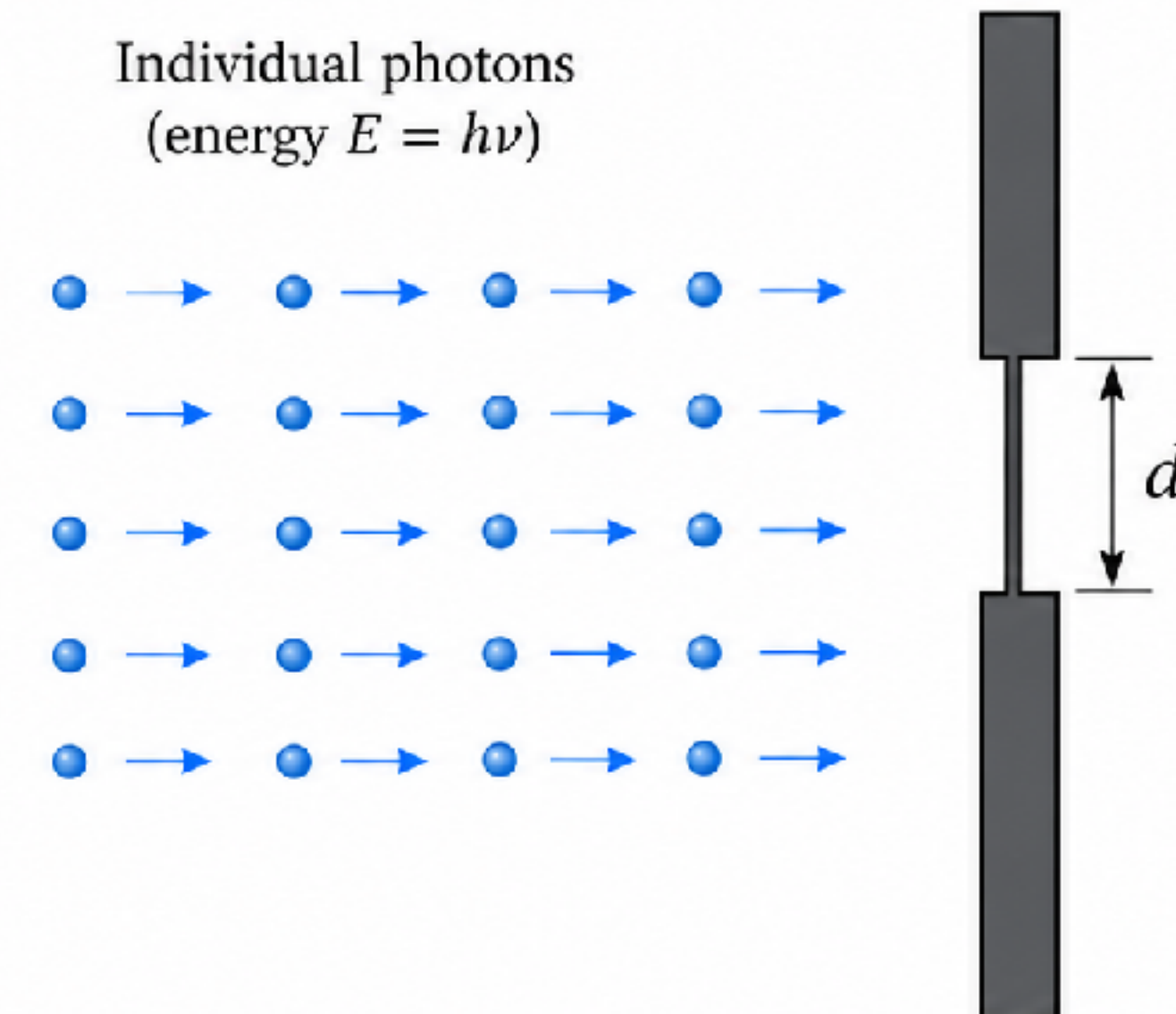
Question: If electromagnetic radiation consists of photons, what happens to the photons in a diffraction or interference experiment?

Interference and diffraction are characteristic wave phenomena. Then blackbody radiation, the photoelectric effect and X-ray production forced us to introduce photons.

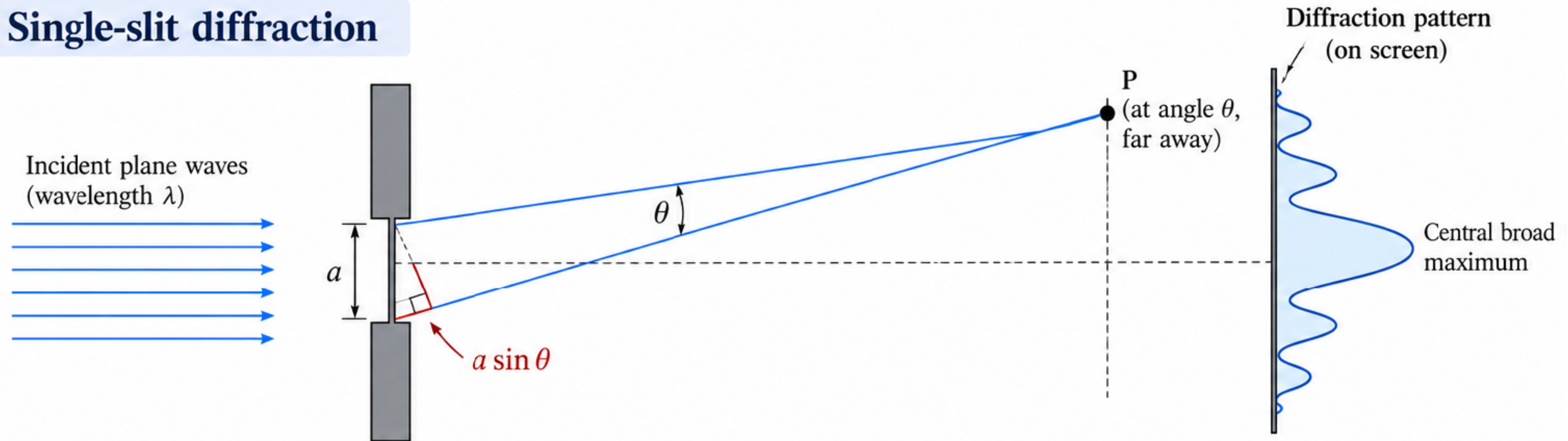
Wave picture



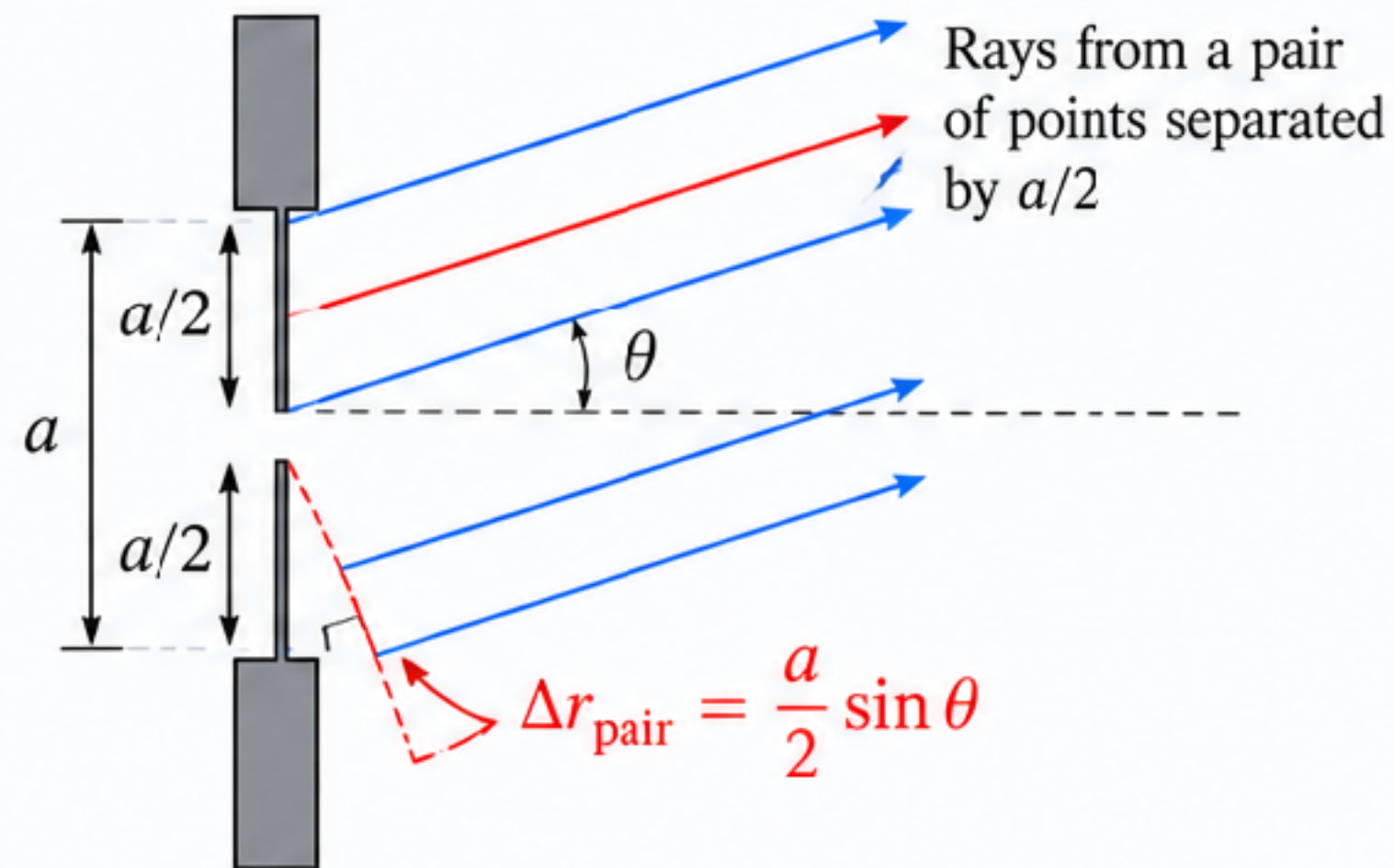
Photon picture



Single-slit diffraction



Pairwise cancellation (for first minimum)



Condition for minima

Path difference within each pair:

$$\Delta r_{\text{pair}} = \frac{a}{2} \sin \theta$$

For destructive interference:

$$\Delta r_{\text{pair}} = \frac{\lambda}{2}$$

Hence,

$$a \sin \theta = \lambda$$

More generally,

$$a \sin \theta = m\lambda \quad (m = 1, 2, 3, \dots)$$

Angular width of central maximum

For the first minimum ($m = 1$):

$$\sin \theta_1 = \frac{\lambda}{a}$$

For small θ ,

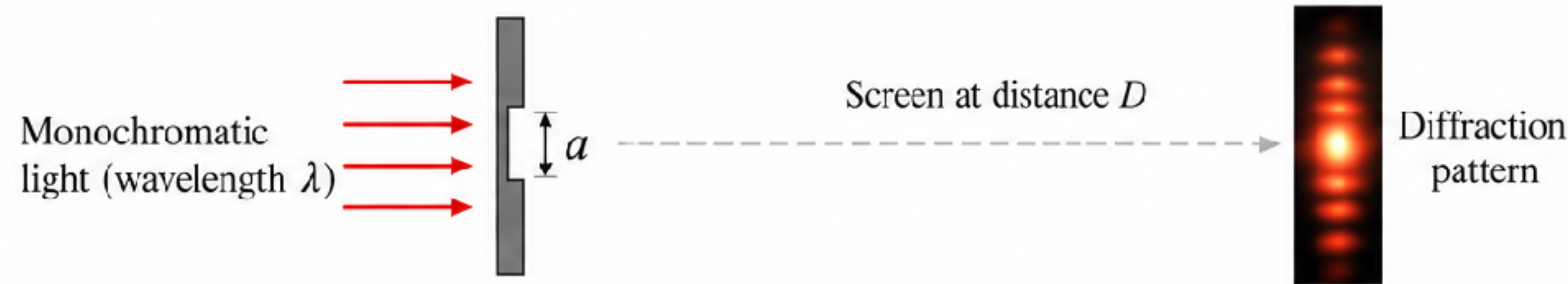
$$\theta_1 \simeq \frac{\lambda}{a}$$

The central maximum extends approximately from $-\frac{\lambda}{a}$ to $+\frac{\lambda}{a}$, so its angular width is

$$\Delta \theta_{\text{central}} \simeq \frac{2\lambda}{a}$$

What controls how much diffraction we see?

Single-slit diffraction: Effect of wavelength λ relative to slit width a



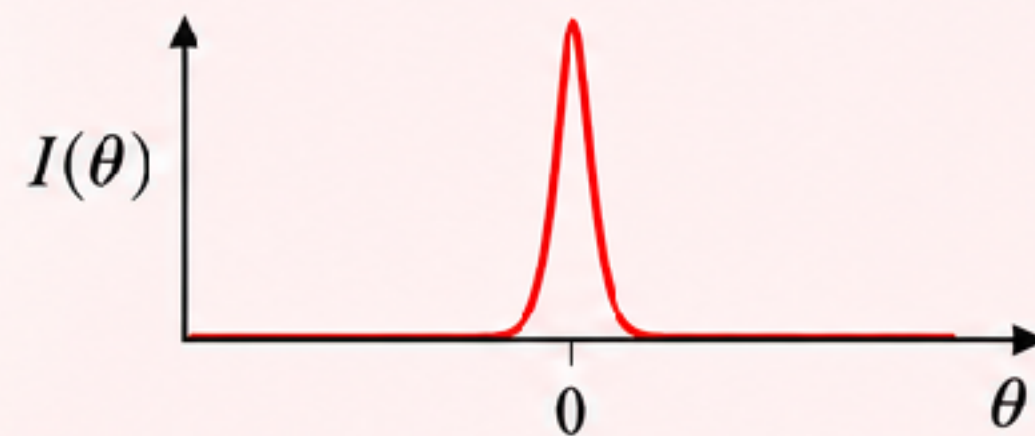
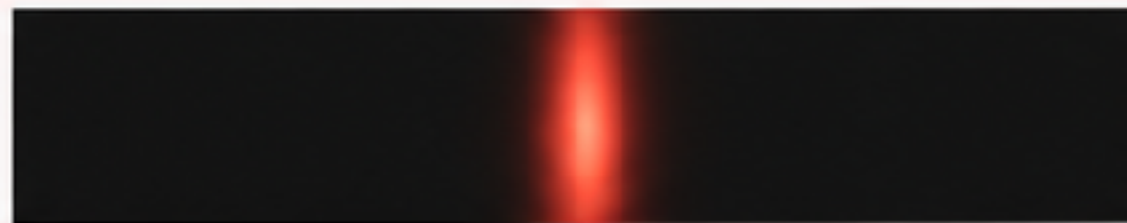
Minima (dark fringes) occur at

$$a \sin \theta = m\lambda \quad (m = \pm 1, \pm 2, \dots)$$

The central maximum ($m = 0$) is at $\theta = 0$.

1. $\lambda \ll a$

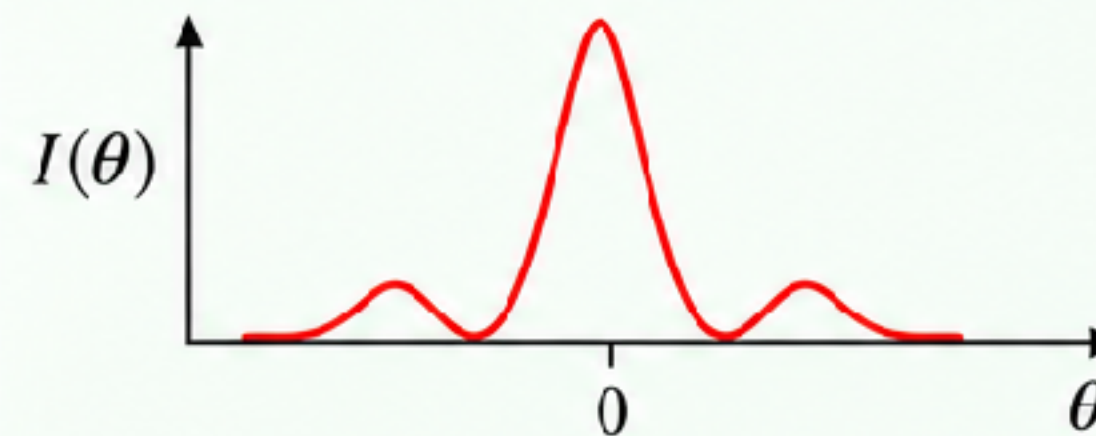
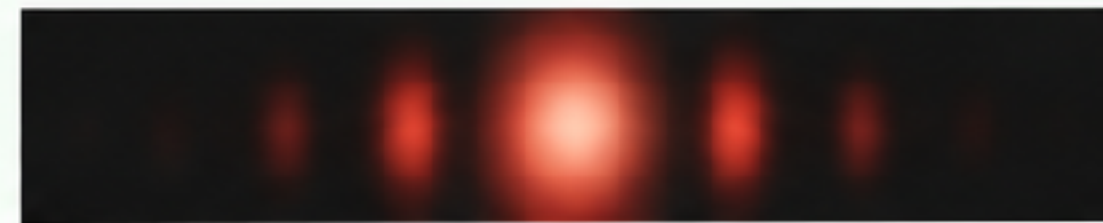
Fringes not resolved (narrow pattern)



- First minimum at $\sin \theta_1 = \frac{\lambda}{a} \ll 1$ (very small angle).
- Fringes are too close to be resolved.
- Diffraction pattern is very narrow, almost like a single central spot.

2. $\lambda \sim a$

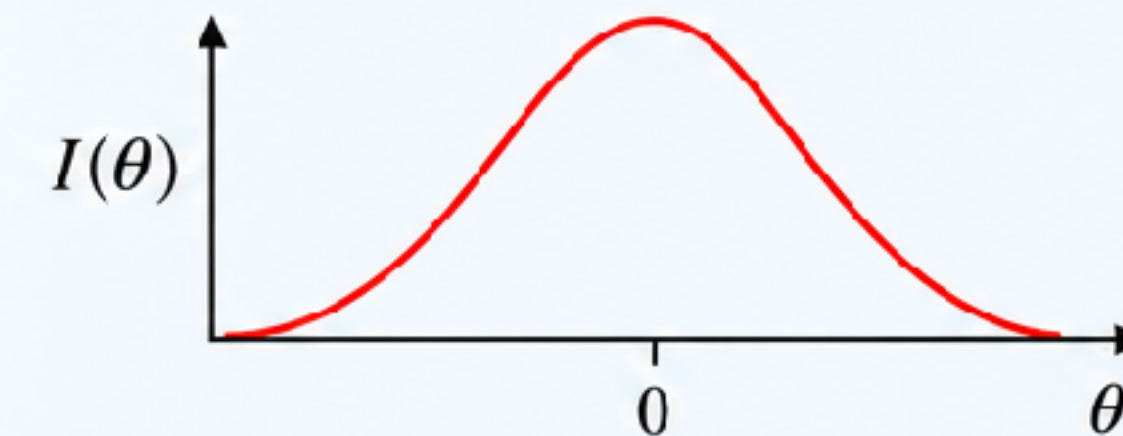
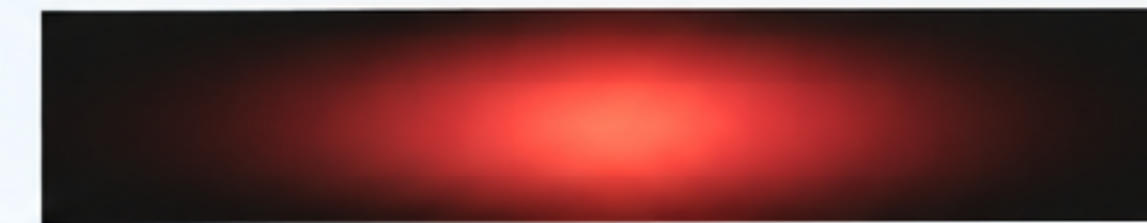
Clear diffraction pattern



- First minimum at $\sin \theta_1 = \frac{\lambda}{a} \sim 1$ (angle of order 1 radian).
- A few side maxima and minima are visible.
- Diffraction pattern is spread over a wide angular range.

3. $\lambda > a$

Only central maximum (no higher orders)



- For $m \geq 1$, $a \sin \theta = m\lambda > \lambda \Rightarrow \sin \theta > 1$ (not possible).
- No higher-order minima or maxima.
- Only a broad central maximum is observed, extending over a large angular range (almost all light goes into the central maximum).

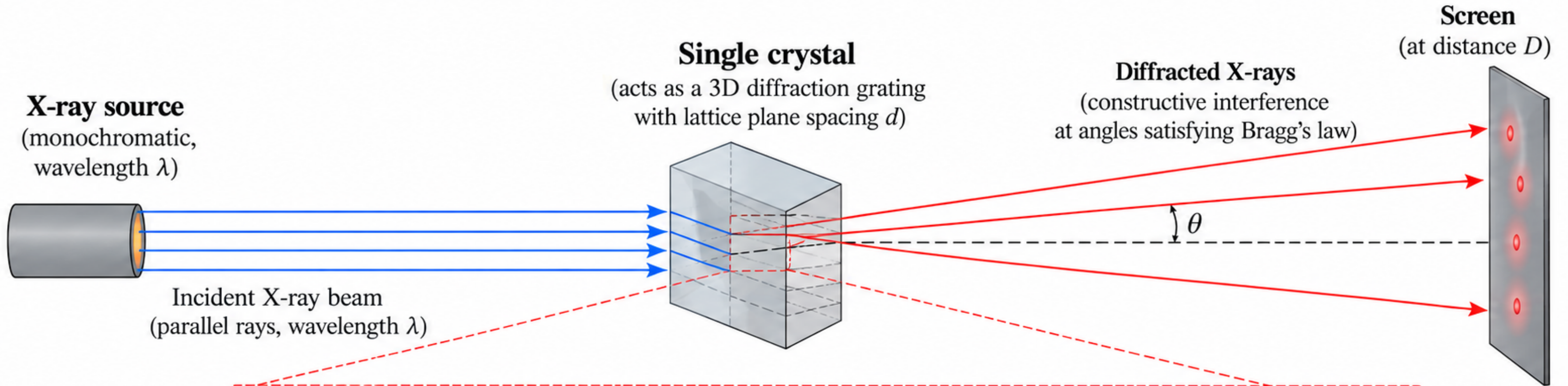
X- ray diffraction

For X-rays, $\nu \approx 10^{19}$ Hz so $\lambda = 0.3 \text{ \AA}$

Nature already gives us a periodic structure on exactly this scale-a crystal.

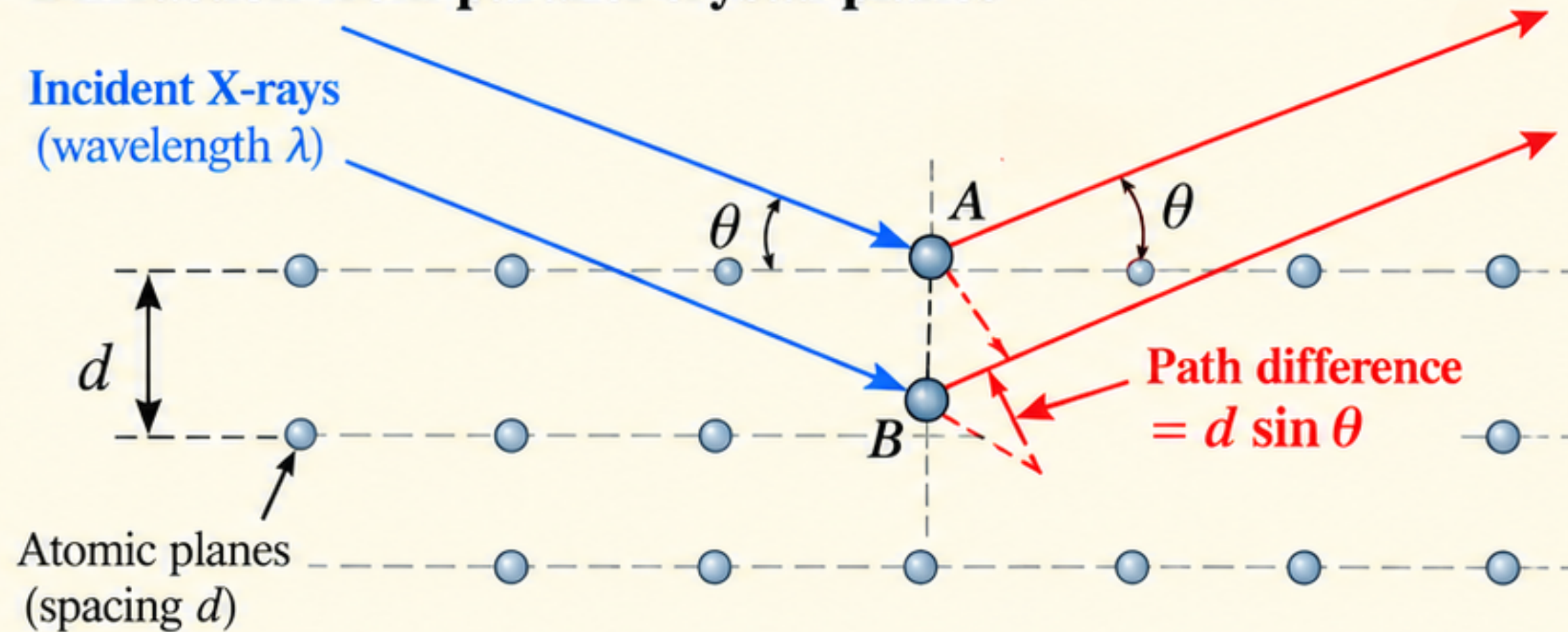
The spacing between atomic planes is of order Ångströms, comparable to X-ray wavelengths. Therefore X-rays scattered from different crystal planes can interfere.

X-ray Diffraction



Diffraction from parallel crystal planes

Incident X-rays
(wavelength λ)



Condition for constructive interference
(Bragg's law)

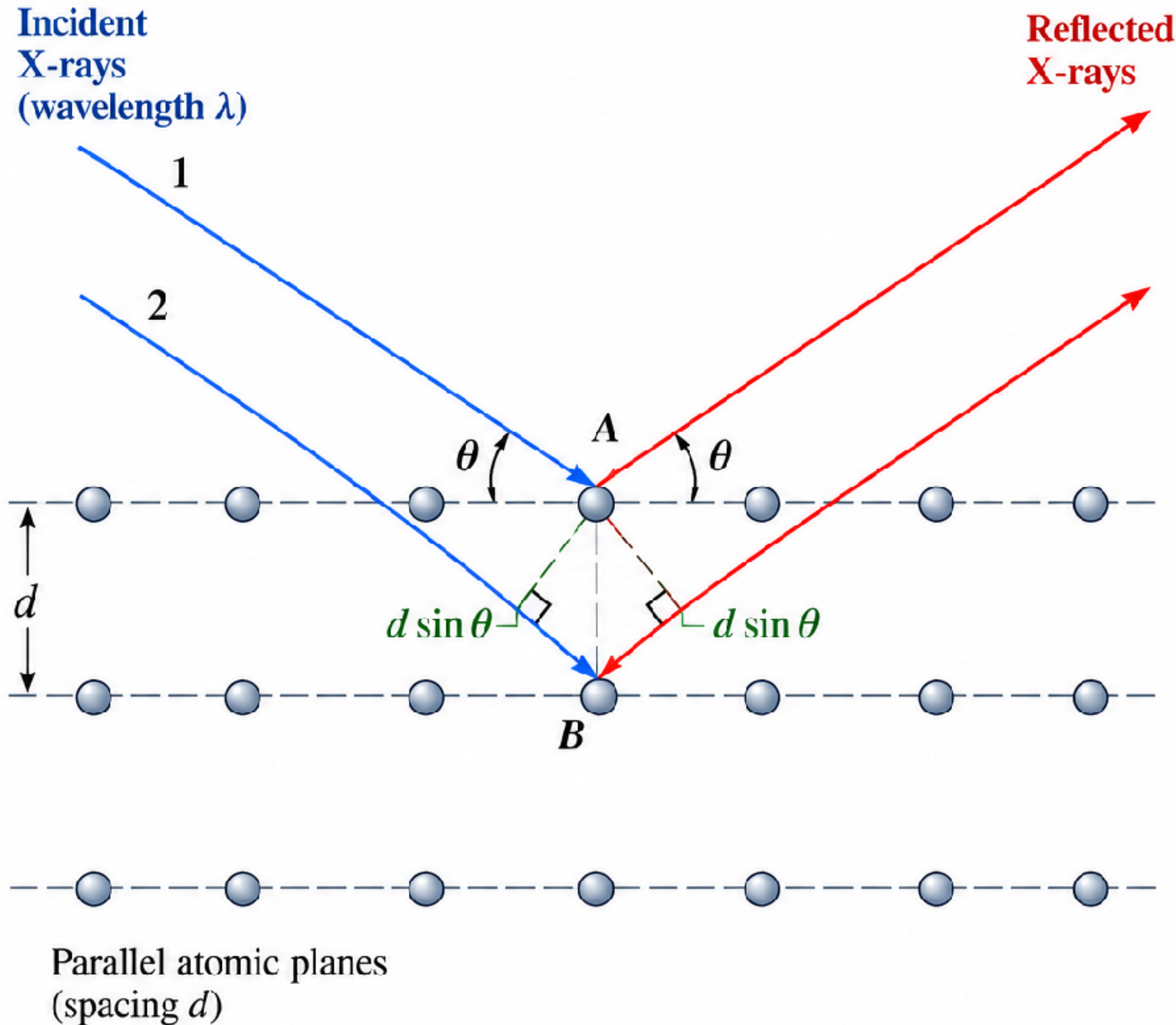
$$d \sin \theta = m\lambda$$

$$m = 1, 2, 3, \dots$$

Rays scattered from successive parallel planes interfere constructively when their path difference is an integer multiple of λ .

X-ray Diffraction

Bragg's law: Diffraction from parallel crystal planes



Condition for constructive interference (Bragg's law):

$$2d \sin \theta = m\lambda$$

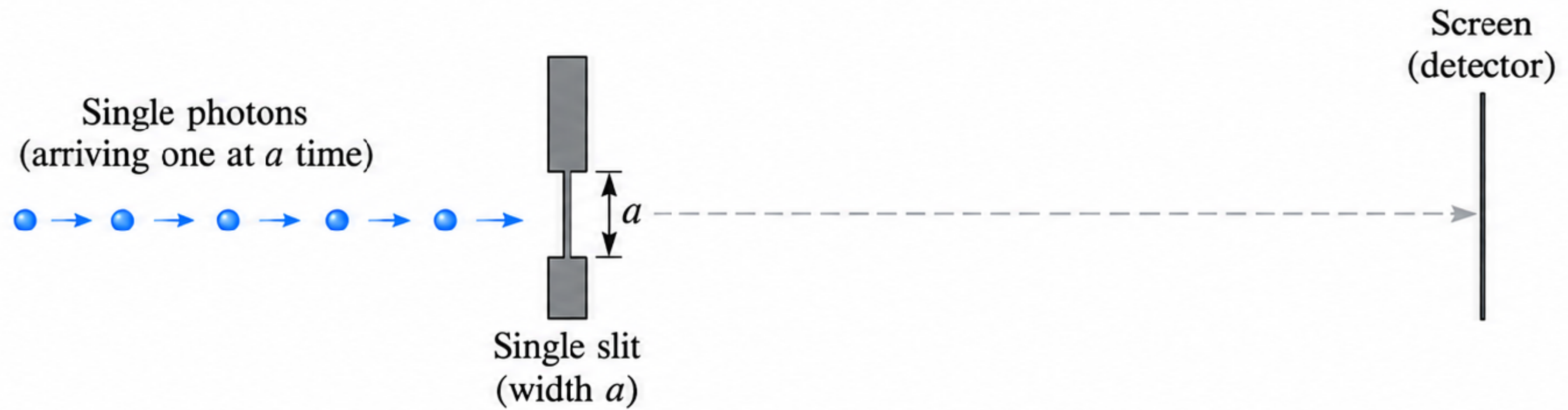
$$m = 1, 2, 3, \dots$$

- Ray 2 travels an additional distance $d \sin \theta$ compared to ray 1 on the way in.
- After reflection, it travels an additional distance $d \sin \theta$ compared to ray 1 on the way out.
- Total path difference = $2d \sin \theta$.
- Constructive interference occurs when $2d \sin \theta = m\lambda$.

Diffraction in Photon picture

Make the X-ray beam extremely weak - Reduce the intensity so much that there are only a few photons such that they cross the slit essentially **one at a time**.

Diffraction in Photon picture



Diffraction in Photon picture

1 photon



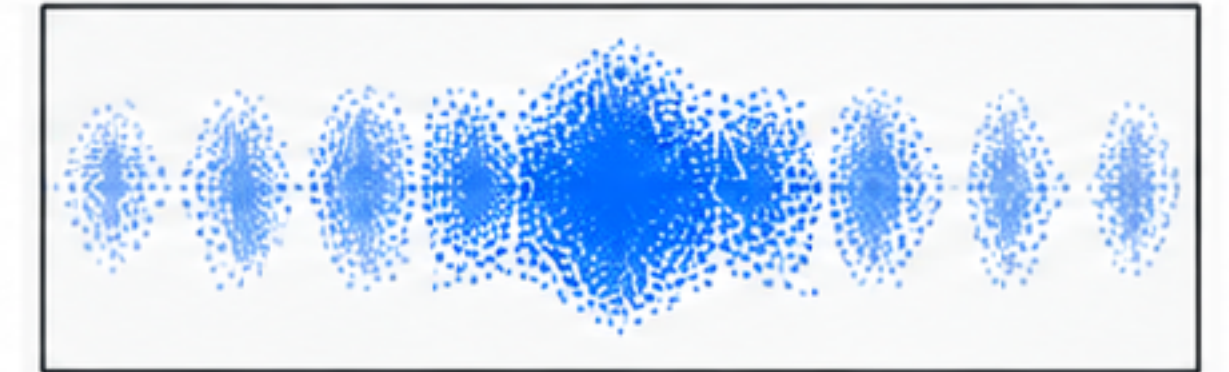
20 photons



500 photons



Many photons



Time (increasing number of detected photons) →

- The first photon arrives at one particular point on the screen.
- The second photon arrives - could be at same location or another point.
- Each detection is localized - it looks completely particle-like.
- And one photon does **not** tell us that a diffraction pattern exists.
- But now repeat the experiment many times.

Diffraction in Photon picture

- We cannot predict exactly where an individual photon will be detected, but different positions do not have equal probabilities.
- The collection of individual photon detections reproduces exactly the diffraction pattern predicted by wave optics.
- So the wave description has not disappeared when we introduce photons. It determines the statistical distribution of photon detections.

